AORTIC ANEURYSM & DISSECTION

OVERVIEW: ANATOMY, ANEURYSM, AND DISSECTION

A weakened or torn aorta—the vessel either balloons silently (aneurysm) or splits along its wall (dissection), and both can rupture into sudden death.

NORMAL AORTIC ANATOMY

Ascending aorta
Rises from the left ventricle.

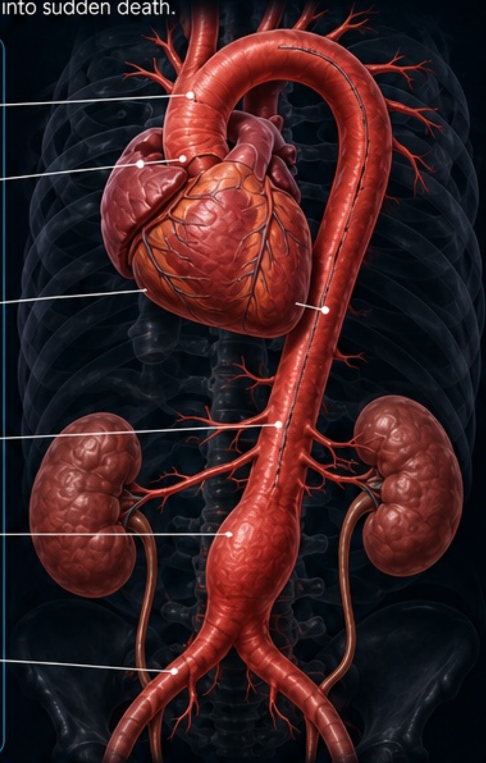
Aortic root
Includes the sinuses of Valsalva; critical in valve function.

Aortic arch
Gives rise to the major branches to the head, neck, and upper limbs.

Descending thoracic aorta
Supplies the chest wall and thoracic organs.

Abdominal aorta
Supplies abdominal organs and lower extremities.

Common iliac arteries
Bifurcation of the aorta to the pelvis and legs.



AORTIC ANEURYSM

Pathologic dilation >50% of normal vessel diameter from medial wall weakening.

Most common in the infrarenal abdominal aorta (AAA).
Often silent until rupture.

AORTIC DISSECTION

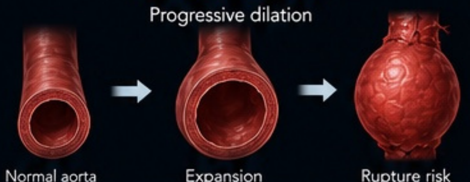
An intimal tear allows blood to enter the media, creating a false lumen.

The false lumen can propagate along the vessel, compromising branch flow and risking rupture.

ANEURYSM VS. DISSECTION: TWO DIFFERENT ENTITIES

AORTIC ANEURYSM

Progressive dilation



Enlarges over years. Often asymptomatic until complications occur.

AORTIC DISSECTION

Intimal tear

Occurs in seconds to minutes. Always a potentially fatal emergency.

STANFORD CLASSIFICATION (DISSECTION)

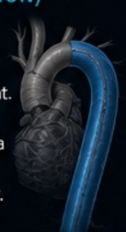
TYPE A

Any involvement of the ascending aorta (± arch or descending).

EMERGENT SURGERY
Surgical emergency.

TYPE B

No ascending aorta involvement. Limited to the descending aorta distal to the left subclavian artery.



MEDICAL MANAGEMENT
Unless complicated.

KEY DRIVERS

- **Uncontrolled hypertension** – Most important modifiable risk.
- **Smoking** – Strongest risk factor for AAA expansion/rupture.
- **Increasing age** – Risk rises with age (>65 years).
- **Atherosclerosis** – Major contributor to AAA.
- **Connective tissue disease** – Marfan, Loeys-Dietz, Ehlers-Danlos.
- **Bicuspid aortic valve** – Associated with aneurysm and dissection.
- **Pregnancy** – Increased risk, especially in connective tissue disease.
- **Cocaine use** – Acute웹 hypertension and aortic injury.
- **Trauma** – Blunt or penetrating injury to the aorta.

KEY POINT | *Aneurysms enlarge over years. Dissections occur in seconds. Recognizing the difference saves lives.*

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Overview: Anatomy, Aneurysm, and Dissection	Overview	Illustration	ACC/AHA Aortic Disease Guidelines (2022) UpToDate (Aortic Aneurysm; Aortic Dissection)	CLINICAL PENDING

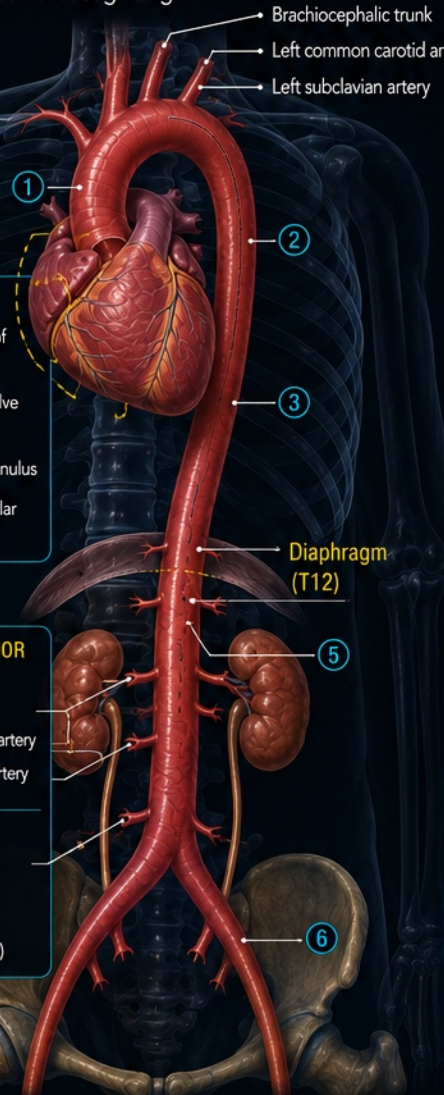
AORTIC ANEURYSM & DISSECTION

LABELED AORTIC ANATOMY

Understanding normal aortic anatomy is essential for recognizing aneurysmal dilation and dissection patterns.

KEY ANATOMY

1. **Ascending aorta**
   Arises from the left ventricle; includes the aortic root.
2. **Aortic arch**
   Curves posteriorly and gives rise to the major branches supplying the head, neck, and upper limbs.
3. **Descending thoracic aorta**
   Runs through the posterior mediastinum; supplies the chest wall and thoracic organs.
4. **Diaphragmatic hiatus (T12)**
   Aorta passes through the aortic hiatus at the level of T12.
5. **Abdominal aorta**
   Supplies abdominal organs and gives rise to unpaired and paired branches.
6. **Common iliac arteries**
   Bifurcation usually occurs near the L4 vertebral level into the right and left common iliac arteries.



AORTIC ROOT

UNPAIRED ANTERIOR BRANCHES

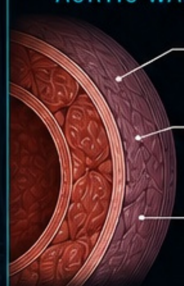
- Celiac trunk
- Superior mesenteric artery
- Inferior mesenteric artery

PAIRED LATERAL BRANCHES

- Renal arteries
- Gonadal arteries (testicular/ovarian)

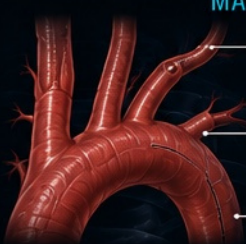
AORTIC SEGMENTS

AORTIC WALL LAYERS



The media is the thickest layer and provides strength and elasticity. It is the layer most affected in aneurysm and dissection.

MAJOR ARCH BRANCHES



1. **Brachiocephalic trunk**
   Supplies right side of the head, neck, and right upper extremity.
2. **Left common carotid artery**
   Supplies left side of the head and neck.
3. **Left subclavian artery**
   Supplies left upper extremity.

Branch	Supplies
Brachiocephalic trunk	Right common carotid artery + Right subclavian artery
Left common carotid artery	Left head and neck
Left subclavian artery	Left upper extremity

KEY POINT | *Know the normal aortic anatomy. It is the roadmap for recognizing abnormal dilation, intimal tears, and branch-vessel compromise.*

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Labeled Aortic Anatomy	Anatomy	Illustration	Netter's Atlas of Human Anatomy (7th Ed.) Rohen's Anatomy (9th Ed.)	CLINICAL PENDING

AORTIC ANEURYSM & DISSECTION

PATHOPHYSIOLOGY

Two distinct disease processes weaken or separate the aortic wall: aneurysms enlarge over years, while dissections occur in seconds.

ANEURYSM: WHY IT ENLARGES

ABDOMINAL AORTIC ANEURYSM (AAA)

- Smoking
- Increasing age
- Atherosclerosis
- Chronic inflammation
- Proteolysis (MMPs, elastin loss)

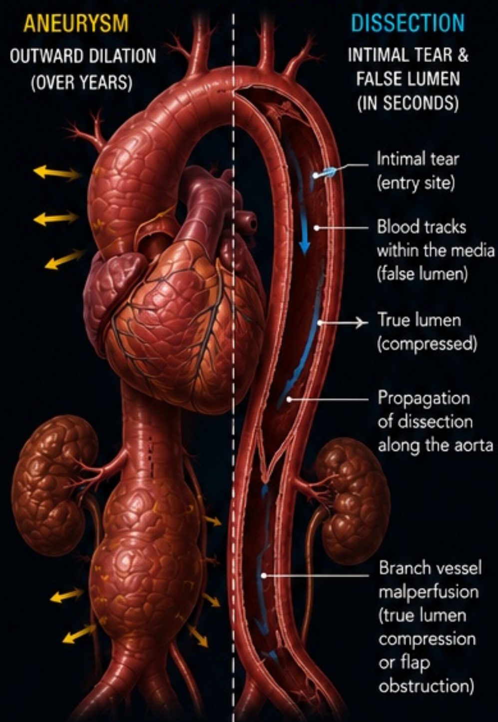
KEY CONCEPT
True aneurysm = permanent dilation involving all three vessel-wall layers.

THORACIC AORTIC ANEURYSM

- Hypertension
- Bicuspid aortic valve
- Heritable aortopathies (Marfan, Loeys–Dietz, EDS)
- Cystic medial degeneration (smooth muscle cell loss, elastin fragmentation)

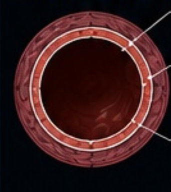
KEY CONCEPT
Results from medial degeneration or genetic/connective tissue disease rather than atherosclerosis.

OVERVIEW OF PATHOPHYSIOLOGY



AORTIC WALL CROSS-SECTIONS

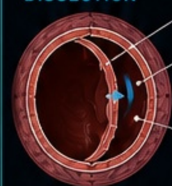
NORMAL AORTA



ANEURYSM (TRUE ANEURYSM)

Outward dilation involving all wall layers, with remodeling and loss of structural integrity.

DISSECTION



Blood dissects within the medial layer of the aortic wall.

PATHOPHYSIOLOGY PATHWAYS

ANEURYSM PATHWAY (YEARS)

1. **Wall degeneration** — Inflammation and matrix degradation weaken the media.

2. **Matrix loss & remodeling** — Elastin fragmentation and smooth muscle cell loss.

3. **Progressive dilation** — Outward expansion of the aorta.

4. **Increasing wall stress** — Higher tension with larger radius and thinner wall.

DISSECTION PATHWAY (SECONDS TO MINUTES)

1. **Susceptible media + hemodynamic stress** — Hypertension or acute pressure surge on a weakened media.

2. **Intimal tear** — A tear allows blood to enter the medial layer.

3. **False lumen creation** — Blood tracks within the media, creating a false lumen.

4. **Propagation** — Dissection extends proximally and/or distally.

5. **Complications** — Branch vessel malperfusion, rupture, or death.

MAJOR RISK FACTORS

MORE COMMON FOR AAA	MORE ASSOCIATED WITH DISSECTION / THORACIC ANEURYSM
Smoking	Hypertension
Older age (≥65)	Bicuspid aortic valve
Atherosclerosis	Heritable connective tissue disease (Marfan, Loeys–Dietz, EDS)
Male sex	Pregnancy (late 2nd–3rd trimester)
	Cocaine or stimulant use
	Family history of aortopathy

HEMODYNAMIC PRINCIPLES

Aortic wall stress rises with higher pressure, larger radius, and thinner wall. Anti-impulse therapy lowers blood pressure, heart rate, and the rate of pressure rise (dP/dt), thereby reducing aortic wall stress and dissection propagation.

SIMPLIFIED CYLINDRICAL-WALL RELATIONSHIP (LAPLACE LAW)

Circumferential wall stress (σ)

$$\boxed{\sigma \approx \frac{P \times r}{2 \times h}}$$

P = Intraluminal pressure
r = Radius of the aorta
h = Wall thickness

This is a simplified relationship to illustrate how wall stress increases.
⚠ Not all dissections or ruptures are predicted solely by size.

KEY POINT

Aneurysms enlarge slowly due to chronic wall degeneration and remodeling. Dissections occur suddenly when an intimal tear allows blood to track within the media. Understanding the mechanism helps anticipate complications and guide treatment.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Pathophysiology of Aortic Aneurysm & Dissection	Pathophysiology	Illustration	ACC/AHA Aortic Disease Guidelines (2022) UpToDate (Aortic Aneurysm; Aortic Dissection)	CLINICAL PENDING

AORTIC ANEURYSM & DISSECTION

CLINICAL PRESENTATION

Symptoms depend on the aortic segment involved, disease process, and the presence of complications such as rupture, dissection extension, or branch vessel malperfusion.

CHRONIC DISEASE (OVER YEARS) — AORTIC ANEURYSM: OFTEN SILENT

VS.

ACUTE VASCULAR EMERGENCY (IN SECONDS) — AORTIC DISSECTION: SUDDEN & SEVERE

- Most aneurysms are asymptomatic until large or complicated.
- Pulsatile abdominal mass (AAA)
- Deep, steady abdominal or back pain
- Early satiety (large AAA)
- Hoarseness, cough, or dysphagia (thoracic aneurysm)

THORACIC ANEURYSM

- Chest, back, or between the shoulder blades
- Cough, dyspnea, or wheezing
- Hoarseness or difficulty swallowing
- Superior vena cava compression (rare)

ABDOMINAL ANEURYSM (AAA)

- Deep abdominal or flank pain
- Pulsatile abdominal mass
- Pulsatile abdominal sensation
- Fullness after eating or early satiety

KEY CONCEPT

True aneurysm = permanent dilation involving all three vessel-wall layers.

Dissection is a medical emergency. Pain is the hallmark in most patients.

PAIN MAXIMAL AT ONSET

CLASSIC PAIN FEATURES

- Sudden onset, "tearing" or "ripping" quality
- Maximal at onset
- May radiate to the back, neck, jaw, or abdomen
- Pain may migrate as the dissection propagates along the aorta
- Migratory pain with extension

OTHER COMMON FINDINGS

- Hypertension (common)
- Hypotension/ shock (rupture or tamponade)
- Pulse deficits or inter-arm systolic BP difference (>20 mmHg)
- Focal neurologic deficit (malperfusion)
- Flank pain, oliguria (renal ischemia)

MALPERFUSION SYNDROMES

Result from extension of dissection or aneurysm-related thrombosis/compression of branch vessels.

CEREBRAL	CORONARY	RENAL	MESENTERIC	LIMB
• Stroke, TIA • Syncope • Confusion	• Chest pain • Myocardial ischemia/MI	• Flank pain • Elevated creatinine • Oliguria/anuria	• Severe abdominal pain • Nausea/vomiting • Bowel ischemia	• Limb pain • Weakness • Cool or pulseless extremity

RUPTURE SIGNS

Aneurysm rupture (most commonly AAA) can be catastrophic. Recognize early.

- SUDDEN SEVERE PAIN
- HYPOTENSION & SHOCK (SBP < 90)
- TACHYCARDIA
- PALLOR & DIAPHORESIS
- DISTENDED, TENDER ABDOMEN

THORACIC RUPTURE
- Hemothorax
- Tamponade

ABDOMINAL RUPTURE
- Retroperitoneal hemorrhage

CLINICAL CLUES THAT INCREASE SUSPICION

- Age > 60 years
- History of hypertension
- Known aortic aneurysm
- Bicuspid aortic valve or connective tissue disorder
- Family history of aortic disease
- Pregnancy (3rd trimester/early postpartum)
- Cocaine or stimulant use
- Sudden unexplained pain (chest, back, abdomen)

PEARL: *Aneurysms often whisper until they rupture. Dissections usually scream from the start. Pain + risk factors + focal deficits = act immediately.*

CONDITION	TYPICAL ONSET	KEY SYMPTOMS	KEY CLUES	RED FLAGS
THORACIC ANEURYSM	Insidious (years)	• Vague chest/back pain • Cough, dyspnea • Hoarseness, dysphagia	• Risk factors for aneurysm • Known dilation	• Rapid enlargement • New/worsening symptoms
ABDOMINAL ANEURYSM (AAA)	Insidious (years)	• Often asymptomatic • Pulsatile abdominal mass • Deep abdominal/back pain	• Age > 65, male, smoker • AAA > 5.5 cm	• Sudden severe pain • Hypotension/shock • Tender or rigid abdomen
AORTIC DISSECTION	Sudden (seconds)	• Tearing/ripping pain • Migratory pain • Neuro or limb symptoms	• Maximal at onset • BP/pulse differential • High-risk conditions	• Hypotension/shock • Neurologic deficit • Signs of malperfusion

WHEN TO SUSPECT: Sudden severe pain → Risk factors present → Focal deficits or pulse/BP difference → Hemodynamic instability → Urgent imaging & treatment → **TIME IS AORTA.** Rapid recognition saves lives.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Clinical Presentation of Aortic Aneurysm & Dissection	Pathophysiology	Illustration	ACC/AHA Aortic Disease Guidelines (2022) UpToDate (Aortic Aneurysm; Aortic Dissection)	CLINICAL PENDING

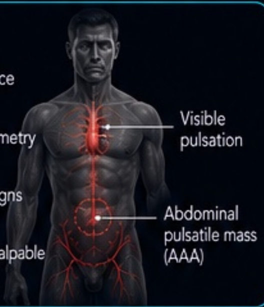
AORTIC ANEURYSM & DISSECTION

PHYSICAL EXAMINATION

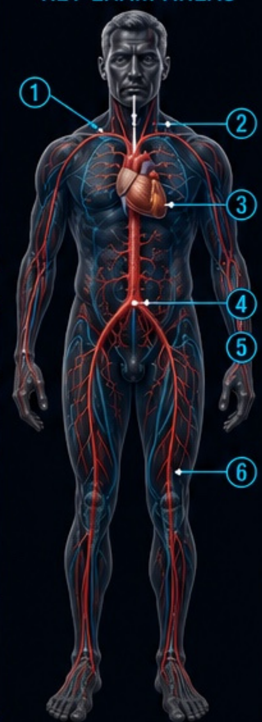
A focused, systematic exam can identify life-threatening aortic disease and complications requiring urgent treatment.

1 INSPECTION

- Assess overall appearance (distress, diaphoresis)
- Look for chest wall asymmetry or visible pulsation
- Check for hypotension signs (pallor, cool extremities)
- Abdominal distension or palpable mass (AAA)



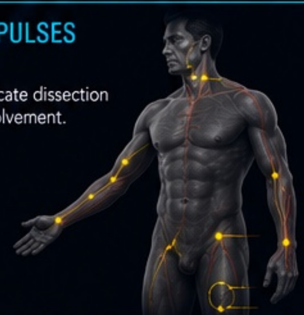
KEY EXAM AREAS



2 PALPATE PULSES

Compare bilaterally.
Asymmetry may indicate dissection or branch vessel involvement.

- Carotid
- Brachial
- Radial
- Femoral
- Popliteal
- Dorsalis pedis
- Posterior tibial



Pulse deficit or asymmetric pulses strongly increase suspicion for acute dissection.

AUSCULTATION

Aortic Area

- New diastolic murmur → acute aortic regurgitation (common in Type A dissection)
- New murmur in sudden chest pain = think dissection

Back (between scapulae)

- Aortic regurgitation murmur
- Periaortic bruit (aneurysm)

Abdomen

- Bruit over enlarged or dissected abdominal aorta

KEY CONCEPT
New diastolic murmur in acute chest pain suggests aortic regurgitation from Type A dissection.

BLOOD PRESSURE

- Measure in both arms.
- Systolic difference > 20 mmHg suggests dissection.
- Severe hypertension is common in acute dissection.
- Hypotension suggests rupture, tamponade, or cardiogenic shock.

INTER-ARM SYSTOLIC BP DIFFERENCE

0–10 mmHg	10–20 mmHg	>20 mmHg
Normal	Borderline	Concerning

Supportive finding — not present in all dissections.

ABDOMINAL EXAM

- Palpate for midline pulsatile mass (AAA typically ≥ 3 cm)
- Assess for tenderness
- Auscultate for abdominal aortic bruit
- Check for peritoneal signs (rupture)

A pulsatile, expansile mass is a red flag for abdominal aortic aneurysm.

NEUROLOGIC & PERFUSION

Neurologic Exam

- Cranial nerves, strength, sensory, speech
- Look for stroke or spinal cord ischemia (dissection)
- Check for syncope

Peripheral Perfusion

- Assess capillary refill (< 2 sec normal)
- Check temperature, color, edema
- Look for acute limb ischemia (pain, pallor, pulselessness, paresthesia, paralysis)

FOCAL NEURO DEFICIT

ACUTE LIMB ISCHEMIA

THINK AORTA! *Sudden severe pain + abnormal exam findings (pulse deficit, BP difference, new diastolic murmur, hypotension, focal deficit, or abdominal mass) =* IMMEDIATE ACTION.

KEY EXAM FINDINGS

FINDING	SUGGESTS	CLINICAL SIGNIFICANCE
Pulse deficit / inter-arm BP difference	Aortic dissection	Indicates branch vessel or true lumen compromise
New diastolic murmur	Acute aortic regurgitation	Common in Type A dissection
Abdominal pulsatile mass	AAA	Risk of rupture increases with size
Hypotension / shock	Rupture / tamponade	Life-threatening emergency
Focal neurologic deficit	Dissection (stroke / spinal ischemia)	Urgent imaging and intervention needed
Acute limb ischemia	Branch vessel occlusion	May result from dissection flap

EXAM PEARLS

- A normal exam does NOT exclude aortic disease.
- The combination of history and exam increases diagnostic suspicion.
- When dissection is suspected, never delay definitive imaging or surgical consultation.

EXAM TO ACTION

Focused History → Targeted Physical Exam → High Suspicion Present? → YES → Urgent Imaging (CTA T/A) → Immediate Management & Consultation →

SAVE THE AORTA. SAVE LIVES.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Physical Examination in Aortic Aneurysm & Dissection	Clinical	Illustration / Infographic	ACC/AHA Aortic Disease Guidelines (2022) UpToDate (Aortic Aneurysm; Aortic Dissection)	CLINICAL PENDING

AORTIC ANEURYSM & DISSECTION

DIAGNOSTIC IMAGING

Imaging confirms the diagnosis, defines the extent, identifies complications, and guides urgent management.

IMAGING STRATEGY

- Time is critical. CTA is the most practical first-line test in most stable patients.
- Tailor modality to the patient's stability, renal function, and clinical presentation.
- Use multiplanar and 3D reconstructions to assess extent and branch vessels.
- Always evaluate for complications: rupture, malperfusion, tamponade, and aortic regurgitation.

CHOOSING THE RIGHT MODALITY

CT ANGIOGRAPHY (CTA)
- First-line test for suspected acute aortic syndrome.
- Fast, widely available, excellent spatial resolution.
- Best for evaluating branch vessels and complications.

TEE (TRANSESOPHAGEAL ECHO)
- Excellent for proximal aorta (root, ascending, arch).
- Ideal for unstable patients or when CTA is contraindicated.
- Portions of the distal ascending aorta and arch may be less well visualized.

MRI / MRA
- Highly accurate for both acute and chronic aortic disease.
- Best for serial surveillance (no ionizing radiation).
- Longer acquisition time; less available emergently.

CONVENTIONAL AORTOGRAPHY
- Invasive; now rarely first-line. Reserved when noninvasive imaging is inconclusive or intervention is planned.

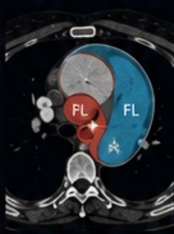
CT ANGIOGRAPHY (CTA): KEY FINDINGS

AORTIC DISSECTION

INTIMAL FLAP

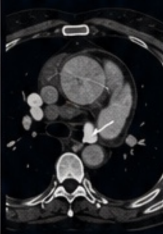
Linear flap within the lumen separating true and false lumens.

TRUE & FALSE LUMEN



True lumen (TL) is the original channel. False lumen (FL) is pressurized.

FLAP ENTRY TEAR



Disruption in the intima allows blood to enter the media.

EXTENT & PROPAGATION

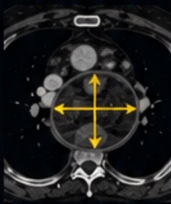
Dissection may extend proximally or distally along the aorta.

BRANCH VESSEL MALPERFUSION

Compression or occlusion of branch vessels may cause ischemia.

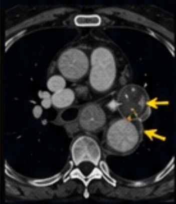
AORTIC ANEURYSM

FUSIFORM ANEURYSM



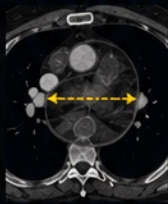
Circumferential dilation involving the entire aortic circumference.

SACCULAR ANEURYSM



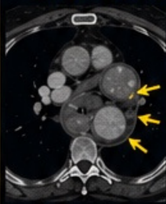
Focal outpouching involving part of the aortic wall.

MAXIMUM DIAMETER



Measure inner-edge to inner-edge (double oblique, perpendicular to centerline).

MURAL THROMBUS

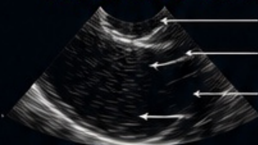


Mural thrombus is common in abdominal aortic aneurysm.

RUPTURE / LEAK

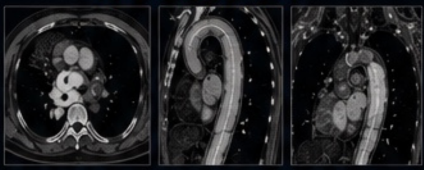
Contrast extravasation, periaortic or retroperitoneal hematoma, or contained rupture.

TEE: KEY VIEW (PROXIMAL AORTA)



Excellent for the aortic root, ascending aorta, valve, pericardial effusion, and proximal descending aorta; portions of the arch may be less well visualized.

MRI / MRA EXAMPLE



Excellent for defining aortic dimensions, branch vessels, and complications without radiation exposure.

HOW TO MEASURE AORTIC DIAMETER

1. Ascending aorta (at level of RPA)
2. Aortic arch (outer curvature)
3. Descending thoracic aorta (at diaphragm)
4. Abdominal aorta (max diameter)

Double oblique measurement perpendicular to centerline

- Measure inner-edge to inner-edge (double oblique, perpendicular to centerline).
- Report in mm and track change over time.

WHEN TO SUSPECT COMPLICATIONS ON IMAGING

- Periaortic hematoma (hyperdense crescent)
- Contrast extravasation (active leak)
- Hemopericardium (tamponade)
- Pleural effusion / hemothorax
- Acute aortic regurgitation (proximal dissection)
- Organ or limb ischemia (malperfusion)

IMAGING PEARLS

- CTA is the fastest and most informative test for acute aortic syndrome.
- Always review axial, sagittal, and coronal images.
- Measure diameters consistently and document the location.
- Look beyond the aorta: search for malperfusion or rupture.
- Early diagnosis and activation of the aortic team saves lives.

DON'T MISS IT! *No single test is perfect. If suspicion remains high despite a negative or indeterminate study, repeat imaging or use an alternative modality.*

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Diagnostic Imaging of Aortic Aneurysm & Dissection	Diagnostic Imaging	Illustration / CT Examples	ACC/AHA Aortic Disease Guidelines (2022) UpToDate (Aortic Aneurysm; Aortic Dissection)	CLINICAL PENDING

AORTIC ANEURYSM & DISSECTION

MANAGEMENT OVERVIEW

A focused approach based on diagnosis, anatomy, and patient stability.

INITIAL STABILIZATION

- Continuous ECG + arterial BP monitoring
- Analgesia for pain control
- IV access, labs, type & screen
- Control blood pressure (SBP < 120 mmHg if tolerated)
- Beta-blocker FIRST-line (esmolol or labetalol titrated to HR < 60 bpm)
- Add vasodilator AFTER beta-blocker if needed (nicardipine or nitroprusside)

⚠ Avoid anticoagulation/antiplatelets unless another diagnosis requires it and dissection has been excluded.

MANAGEMENT BY DIAGNOSIS

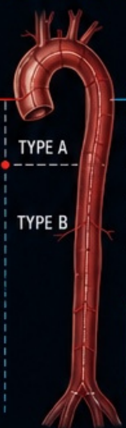
TYPE A (PROXIMAL)
Involves ascending aorta

AORTIC EMERGENCY

- Emergent surgical repair
- IV beta-blocker ± vasodilator
- Urgent surgical consult

↓

OPERATING ROOM
Goal: Prevent rupture, tamponade & end-organ malperfusion



TYPE B (DISTAL)
Distal to left subclavian

ANTI-IMPULSE THERAPY

- Medical management
- IV/PO beta-blocker
- BP control (SBP 100–120 mmHg)
- Close imaging surveillance
- Treat complications promptly

↓

TEVAR IF COMPLICATED
Rupture • Malperfusion
Refractory pain • Rapid expansion

REPAIR THRESHOLDS (ADULTS)

- Ascending aorta / Root ≥ 5.5 cm
- Descending thoracic aorta ~5.5–6.0 cm (individualize based on patient factors)
- Abdominal aortic aneurysm (AAA) ≥ 5.5 cm in men; ≥ 5.0 cm in women
- Rapid growth ≥ 0.5 cm in 6 months

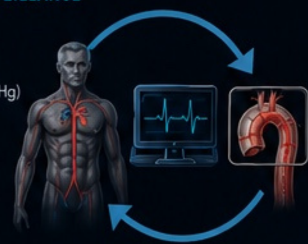
★ Lower thresholds if connective tissue disorder, family history, saccular aneurysm, or other high-risk features.

SURGICAL / ENDOVASCULAR OPTIONS

Procedure	Indication	Key Points
Open Surgical Repair	• Type A dissection • Ascending aortic aneurysm	• Gold standard for Type A • Requires cardiopulmonary bypass
Endovascular Repair (TEVAR)	• Type B dissection • Descending thoracic aneurysm	• Less invasive • Not for ascending aorta
EVAR (Endovascular AAA Repair)	• Abdominal aortic aneurysm	• Preferred for suitable anatomy • Lower perioperative risk
Hybrid / Open Approach	• Complex anatomy • Arch involvement	• Tailored to patient anatomy • Consider comorbidities

FOLLOW-UP & SURVEILLANCE

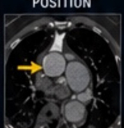
- Imaging at discharge and at intervals
- Strict BP control (target SBP 100–120 mmHg)
- Lifestyle modification (stop smoking, exercise, weight control)
- Long-term surveillance for aneurysm growth or complications
- Genetic evaluation and family screening when appropriate



⚠ Lifelong follow-up is essential

POST-REPAIR IMAGING (CHECK FOR COMPLICATIONS)

GRAFT / STENT POSITION

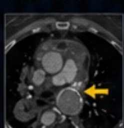


Proper graft / stent position and apposition

ENDOLEAK (EVAR)

Contrast outside graft (Type I, II, or III endoleak)

PSEUDOANEURYSM



Luminal outpouching at repair site

DISSECTION RESIDUAL

Persistent dissection or false lumen

PERFUSION TO BRANCHES

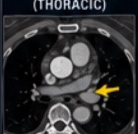
Adequate perfusion to branch vessels (renal, visceral, iliac)

IMAGING MODALITY QUICK REFERENCE

Modality	Best For	Key Advantages	Limitations
CTA	Diagnosis, acute aortic syndromes	Fast, widely available, high resolution	Radiation, iodinated contrast
TEE	Proximal aorta, unstable patients	Bedside, no radiation, excellent root/arch eval	Limited view of arch/distal aorta
MRI / MRA	Chronic follow-up, complex anatomy	No radiation, excellent soft tissue contrast	Time-consuming, limited in unstable patients

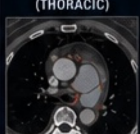
AORTIC ANEURYSM TYPES (CT EXAMPLES)

ASCENDING (THORACIC)



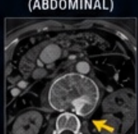
Aneurysm of ascending aorta near the heart

DESCENDING (THORACIC)



Aneurysm of descending thoracic aorta

INFRARENAL (ABDOMINAL)



Infrarenal abdominal aortic aneurysm

JUXTARENAL (ABDOMINAL)

Juxtarenal abdominal aortic aneurysm

MEASUREMENT GUIDELINES (CT / MRI)

- Measure PERPENDICULAR to the vessel centerline
- Use INNER-EDGE to INNER-EDGE ("leading edge" to "leading edge")
- Use ECG-gated imaging for aortic root and ascending aorta
- Use OUTER-EDGE to OUTER-EDGE if wall disease is present
- Report at the standard landmarks

Measure Perpendicular (Double Oblique)

Inner Edge to Inner Edge (Preferred)

Outer Edge to Outer Edge (If Wall Disease)

KEY COMPLICATIONS TO MONITOR (LOCATION MATTERS)

- Acute Aortic Regurgitation
- Coronary Malperfusion
- Stroke / Cerebral Ischemia
- Mesenteric Ischemia
- Renal Ischemia
- Limb Ischemia
- Spinal Cord Ischemia

⚠ RED FLAGS (URGENT ACTION)

- Sudden, severe chest, back, or abdominal pain
- Pulse deficit or asymmetric pulses
- New neurologic deficit or syncope
- Hemodynamic instability or signs of shock
- Evidence of rupture or end-organ malpererfusion

KEY TAKE-HOME
Early recognition, appropriate imaging, and rapid management are critical to preventing catastrophic complications and improving outcomes.

DON'T DELAY
When in doubt, image and act.

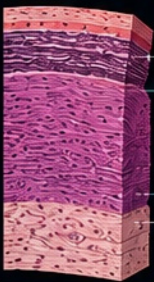
SAVE THE AORTA. SAVE LIVES.

AORTIC ANEURYSM & DISSECTION

HISTOPATHOLOGY & MOLECULAR PATHOGENESIS

Different mechanisms contribute to thoracic aortic aneurysm & dissection versus abdominal aortic aneurysm.

NORMAL AORTIC WALL (THORACIC)

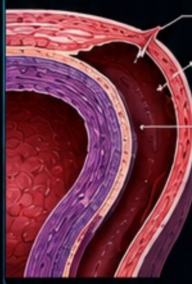


Intima
Endothelium + subendothelial CT; internal elastic lamina (IEL)

Media
Concentric elastic lamellae with smooth muscle cells (SMCs) and proteoglycans

Adventitia
Collagen-rich CT with vasa vasorum and nervi vasorum

AORTIC DISSECTION: INTIMAL TEAR (TYPICAL)



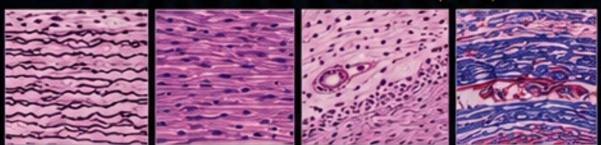
Blood enters through the intimal tear and creates a false lumen that propagates within the media.

ABDOMINAL AORTIC ANEURYSM (AAA)

Transmural inflammatory degeneration with degradation of elastin and collagen, leading to progressive dilation and weakening of the aortic wall.

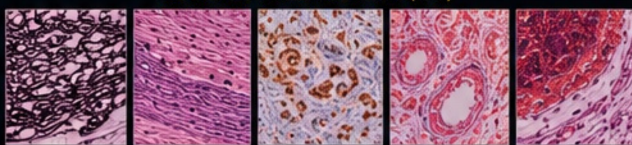
MICROSCOPIC FEATURES

THORACIC AORTIC ANEURYSM / DISSECTION (MEDIA)



Medial degeneration	Smooth muscle cell loss	Vasa vasorum	Collagen remodeling
Fragmentation and loss of elastic lamellae with mucoid matrix accumulation.	Decreased SMCs with apoptosis and replacement by ground substance.	Neovascularization of the media and adventitia (predisposes to rupture).	Disorganized collagen deposition and increased collagen content.

ABDOMINAL AORTIC ANEURYSM (AAA)



Elastin & collagen degradation	Smooth muscle cell depletion	Chronic inflammation	Neovascularization	Mural thrombus (common in AAA)
Loss and fragmentation of elastic fibers and collagen.	Thinning of media with marked loss of SMCs.	Macrophages, T cells and lymphocytes infiltrate the media and adventitia.	Increased vasa vasorum within the adventitia.	Intraluminal thrombus releases proteases and cytokines that promote wall injury.

MOLECULAR MECHANISMS

- Imbalance between proteolytic enzymes (MMPs) and their inhibitors (TIMPs) promotes ECM degradation.
- MMPs (MMP-2, MMP-9, others) degrade elastin and collagen.
- Oxidative stress and reactive oxygen species activate inflammatory and proteolytic pathways.
- Transforming growth factor-β (TGF-β) dysregulation contributes in heritable thoracic aortopathies.
- Smooth muscle cell apoptosis and phenotypic switching weaken the media.
- Extracellular matrix remodeling leads to loss of tensile strength and progressive dilation.

Proteolytic Enzymes > TIMP (Inhibitors) = Matrix degradation & wall weakening

GENETIC & CONNECTIVE TISSUE DISORDERS

- Marfan syndrome (FBN1 mutation)
- Loeys–Dietz syndrome (TGFBR1/2, SMAD2/3, TGFB2/3)
- Vascular Ehlers–Danlos syndrome (COL3A1)
- Turner syndrome
- Bicuspid aortic valve (BAV) associated aortopathy
- Familial thoracic aortic aneurysm/dissection (ACTA2, MYH11, MYLK, LOX, PRKG1 and others)

Genetic factors weaken the aortic wall and increase susceptibility to dilation and dissection.

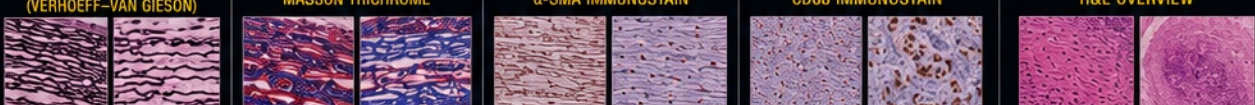
ACQUIRED RISK FACTORS

- Hypertension (most important)
- Cigarette smoking
- Atherosclerosis
- Advanced age
- Chronic inflammation
- Matrix degeneration

KEY PEARLS

- Thoracic aortic disease is primarily due to medial degeneration and heritable or hemodynamic stress.
- AAA is driven by chronic inflammation, SMC loss, and ECM degradation.
- Genetics play a major role in early-onset and aggressive aortic disease.
- Inflammation and MMP activation accelerate wall destruction.
- Early detection and risk factor control slow progression and prevent catastrophe.

SPECIAL STAINS – AORTIC MEDIA



ELASTIN STAIN (VERHOEFF–VAN GIESON)	MASSON TRICHROME	α-SMA IMMUNOSTAIN	CD68 IMMUNOSTAIN	H&E OVERVIEW
Normal / AAA / Dissection	Normal / AAA / Dissection	Normal / AAA / Dissection	Normal / AAA / Dissection	Normal / AAA / Dissection
Loss and fragmentation of elastic fibers in disease.	Increased collagen (blue) with disorganized architecture.	Reduced smooth muscle cell content in diseased media.	Increased macrophage infiltration in aneurysm and dissection.	Overview of wall architecture and inflammatory changes.

STRUCTURE-TO-FAILURE SEQUENCE

Genetic susceptibility or risk factors → Smooth muscle cell dysfunction/loss → Matrix degradation (elastin & collagen) → Aneurysm formation or dissection → Rupture or malperfusion (catastrophic outcomes)

THE TAKEAWAY

Thoracic aneurysm and dissection commonly reflect medial degeneration and heritable or hemodynamic stress, whereas AAA is characterized by inflammation, smooth muscle loss, and extracellular matrix destruction.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Histopathology & Molecular Pathogenesis	Pathology	Illustration / Histology	Robbins Basic Pathology (10e) UpToDate (Aortic Aneurysm; Aortic Dissection)	CLINICAL PENDING

AORTIC ANEURYSM & DISSECTION

TREATMENT & PROCEDURAL MANAGEMENT

Choosing the optimal strategy based on location, severity, anatomy, and patient factors.

TREATMENT PRINCIPLES

- Prevent rupture or malperfusion
- Control blood pressure and reduce shear stress
- Address complications and end-organ ischemia
- Lifelong surveillance and risk factor modification

TREATMENT PARADIGM

TYPE A DISSECTION
(Involves Ascending Aorta)

EMERGENT SURGICAL REPAIR
Goal: Prevent rupture, tamponade, and malperfusion

TYPE B DISSECTION
(Distal to Left Subclavian)

INITIAL MEDICAL MANAGEMENT
Anti-impulse therapy and close monitoring

INTERVENE (TEVAR) IF COMPLICATED
- Rupture
- Malperfusion or branch vessel occlusion
- Extension
- Progressive enlargement
- Intractable pain
- Uncontrolled hypertension

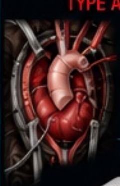
ABDOMINAL AORTIC ANEURYSM (AAA)

- Repair when indicated based on symptoms, size, growth rate, and patient factors
- Shared decision-making based on anatomy, operative risk, life expectancy, and ability to comply with surveillance
- Endovascular or open repair selected based on individual suitability

PROCEDURAL OPTIONS

APPROACH	OPEN SURGICAL REPAIR	ENDOVASCULAR REPAIR (TEVAR / EVAR)	HYBRID / BRANCHED / FENESTRATED	OPEN (COMPLEX / EXTENT II+) THORACOABDOMINAL REPAIR
INDICATIONS	• Type A dissection (standard of care) • Ruptured or symptomatic AAA • Anatomy unsuitable for endovascular repair • Young, low-risk patients when durability prioritized	• Type B dissection with complications • Degenerative aneurysms meeting size criteria or with symptoms • High-risk patients when anatomy is suitable for endovascular repair	• Juxtarenal or para-renal AAA • Anatomy not suitable for standard EVAR • Complex thoracoabdominal aneurysms	• Extensive thoracoabdominal aneurysms • Failed endovascular repair • Infection or connective tissue disorders
ANATOMIC SUITABILITY	Can address a wide range of anatomy, including complex, hostile, or infected aorta.	Requires adequate proximal & distal landing zones and device-compatible anatomy.	Used when standard EVAR not possible but anatomy can be treated with custom or branched solutions.	Preferred for extensive disease requiring direct visualization and reconstruction.
TECHNIQUE OVERVIEW	 Open exposure → resection of diseased segment → graft replacement → reimplantation of branch vessels as needed.	 Endovascular access → stent graft deployment → exclusion of aneurysm or false lumen → preservation of branch flow.	 Custom or off-the-shelf grafts with fenestrations or branches maintain flow to visceral and renal arteries.	 Extensive open exposure → aortic replacement with segmental or temporizing perfusion as needed.
KEY ADVANTAGES	• Most durable long-term results • Comprehensive repair in one setting • No device-related endoleak	• Minimally invasive • Lower perioperative morbidity & mortality • Shorter hospital stay and faster recovery	• Extends endovascular options • May reduce physiologic stress in selected patients	• Durable repair • Direct branch revascularization • Definitive solution for complex disease
LIMITATIONS	• Higher perioperative morbidity • Longer recovery and hospitalization	• Lifelong imaging surveillance • Not suitable for all anatomies • Risk of endoleak or device failure	• Requires advanced planning & expertise • Longer procedure time • Branch, access, endoleak, and spinal cord risks	• Highest physiologic stress • Longer ICU stay & recovery
TYPICAL COMPLICATIONS	• Bleeding • Stroke • Spinal cord ischemia • Renal dysfunction • Infection	• Endoleak (Types I–V) • Access site complications • Stent graft migration • Spinal cord ischemia	• Endoleak • Branch vessel occlusion • Access complications • Contrast nephropathy • Spinal cord ischemia	• Spinal cord ischemia • Renal failure • Respiratory complications • Bleeding

TYPE A DISSECTION REPAIR (OPEN)

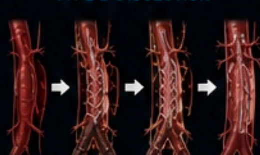


- Replace ascending aorta ± hemiarch/total arch
- Resuspend or replace aortic valve if needed
- Reimplant coronary arteries
- Use of hypothermia & circulatory arrest or antegrade cerebral perfusion for arch repair

GRAFT / CONDUIT OPTIONS

Woven polyester vascular graft (Dacron) | Valve-sparing root reconstruction (if feasible) | Valved conduit (Root or valve replacement needed)

TEVAR FOR COMPLICATED TYPE B DISSECTION



- Cover primary entry tear(s)
- Promote false lumen thrombosis
- Re-expand true lumen
- Confirm branch vessel patency

EVAR FOR AAA

- Adequate proximal & distal landing zones required
- Exclude aneurysm from circulation
- Monitor for endoleak with imaging

ENDOLEAK TYPES

- **Type I** Inadequate seal at proximal or distal attachment site
- **Type II** Retrograde flow from branch vessels (e.g., lumbar, IMA)
- **Type III** Leak due to defect or separation of stent graft
- **Type IV** Porous graft (rare)
- **Type V (Endotension)** Continued aneurysm expansion without visible leak

PERIOPERATIVE MANAGEMENT

- Anti-impulse therapy before repair when indicated (target SBP 100–120 mmHg if tolerated with adequate end-organ perfusion)
- Beta-blockade ± vasodilator therapy
- Optimize cardiac, pulmonary, and renal function
- Spinal cord protection for extensive thoracic/thoracoabdominal coverage when indicated (CSF drainage, MAP augmentation, staging)
- DVT prophylaxis and early mobilization
- Multidisciplinary team approach

LONG-TERM MANAGEMENT

- Lifelong imaging surveillance (CTA or MRA at regular intervals)
- Aggressive risk factor modification (BP control, statin, smoking cessation)
- Screen family members when indicated (genetic & connective tissue disorders)
- Patient education on symptoms requiring urgent evaluation

CLINICAL PEARLS

- Type A dissection is a surgical emergency.
- Symptoms or rupture override size thresholds in AAA.
- EVAR vs open repair is based on anatomy, patient risk, life expectancy, and surveillance compliance.
- Endoleak and device complications require lifelong follow-up.
- Early diagnosis and appropriate management save lives.

RUPTURE CAN BE SUDDEN AND FATAL. EARLY RECOGNITION, RAPID MANAGEMENT, AND LIFELONG FOLLOW-UP ARE ESSENTIAL.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Treatment & Procedural Management	Therapeutics	Illustration / Diagram	2022 ACC/AHA Guideline for the Diagnosis & Management of Aortic Disease	CLINICAL PENDING

AORTIC ANEURYSM & DISSECTION

CLINICAL PEARL & KEY TAKEAWAYS

Early recognition, precise imaging, and appropriate management save lives.

KEY TAKEAWAYS

- Aortic disease is a spectrum—aneurysm, dissection, and rupture—driven by wall degeneration and hemodynamic stress.
- Type A dissections are surgical emergencies.
- Type B dissections are initially treated medically unless complicated.
- AAA repair is based on symptoms, size, growth rate, anatomy, and patient factors.
- Lifelong surveillance is escential after any aortic disease or repair.
- A multidisciplinary approach optimizes outcomes and long-term survival.

RED FLAG PRESENTATIONS

- Sudden, severe, tearing chest, back, or abdominal pain
- Pain migrates (e.g., chest → back, back → abdomen)
- Syncope, near-syncope
- Shortness of breath
- Neurologic deficit or stroke symptoms
- Limb pain, coldness, or pulse deficit
- Hypotension or shock

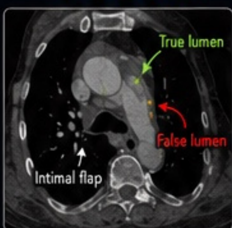
BEDSIDE CLUES

- Pulse or blood pressure differential between arms
- New aortic regurgitation murmur
- Focal neurologic deficit
- Hoarseness or dysphagia (vagus or recurrent laryngeal nerve involvement)
- Signs of malperfusion (renal failure, ischemic bowel, limb ischemia)
- Hypotension or unexplained shock

IMAGING PEARLS

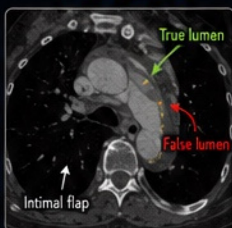
- CTA is the preferred rapid first-line imaging test for most stable patients with suspected acute aortic syndrome.
- Identify intimal flap, true and false lumens, and entry tear.
- Assess branch vessel involvement and organ perfusion.
- Measure perpendicular to the centerline using a consistent, guideline-appropriate edge convention.
- MRI is excellent for surveillance without radiation.
- TEE is rapid and valuable in unstable or non-diagnostic CT.

TYPE A DISSECTION



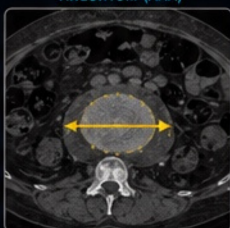
Intimal flap in ascending aorta with false lumen.

TYPE B DISSECTION



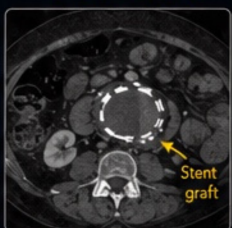
Intimal flap in descending thoracic aorta.

ABDOMINAL AORTIC ANEURYSM (AAA)



Max diameter ≥ 5.5 cm in men (≥ 5.0 cm in women or symptomatic).

POST-EVAR FOLLOW-UP



Exclusion of aneurysm sac with endograft in place.

After EVAR, assess for: *endoleak, graft migration, limb occlusion, aneurysm sac enlargement, and branch vessel perfusion.*

MODIFIABLE RISK FACTORS

- Control hypertension
- Quit smoking
- Treat dyslipidemia
- Maintain healthy weight and regular exercise
- Manage diabetes
- Screen and treat family members when indicated

RAPID SUMMARY: WHAT TO DO

Scenario	First Action	Definitive Strategy
Suspected Type A Dissection	Stabilize, beta-blockade, urgent CTA, consult CT surgery	Emergent surgical repair
Uncomplicated Type B Dissection	Anti-impulse therapy (beta-blocker ± vasodilator), close monitoring	Medical management and surveillance
Complicated Type B Dissection	Stabilize, identify complication, consider endovascular or surgical intervention	TEVAR or surgery based on anatomy and complication
Abdominal Aortic Aneurysm (AAA)	Assess size, symptoms, growth rate, and patient factors	EVAR or open repair when indicated

KEY NUMBERS TO REMEMBER

≥ 5.5 cm (men)
≥ 5.0 cm (women*)
AAA repair threshold. Symptoms or rapid growth may override size.
*Commonly considered lower threshold in women.

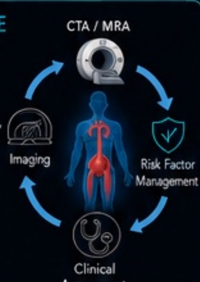
≥ 0.5 cm in 6 months or ≥ 1.0 cm per year
Confirmed rapid growth may prompt repair depending on location and context.

AAA definition: infrarenal aorta ≥ 3.0 cm (screening standard).

Immediate
Type A dissection = operating room.

FOLLOW-UP & SURVEILLANCE

- Surveillance timing depends on diagnosis, repair type, residual disease, anatomy, and prior imaging stability.
- Typical after EVAR: CTA or MRA at 1 month, 6 months, 12 months, then annually if stable (adjust based on endoleak or sac growth).
- Monitor for endoleak, graft integrity, or aneurysm enlargement.
- Aggressive risk factor modification at every visit.
- Genetic counseling and family screening when indicated.



DECISION PATHWAY (SIMPLIFIED)

Suspected Acute Aortic Syndrome
↓
Stabilize Patient (ABC, IV access, monitoring, BP & pain control)
↓
CTA (as indicated) – range guided by clinical scenario & institutional protocol
↓
- Type A Dissection → Emergent Surgical Repair
- Type B Dissection
  - Uncomplicated → Medical Management & Surveillance
  - Complicated → TEVAR or Surgery

FINAL CLINICAL PEARL

> “Time is the greatest determinant of outcome in aortic disease.
> *Suspect early.*
> *Image fast.*
> *Stabilize smart.*
> *Act decisively.*”

RUPTURE CAN BE SUDDEN AND FATAL. EARLY RECOGNITION, RAPID MANAGEMENT, AND LIFELONG FOLLOW-UP ARE ESSENTIAL.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Clinical Pearl & Key Takeaways	Therepeutics	Illustration / Diagram / CT	2022 ACC/AHA Guideline for the Diagnosis & Management of Aortic Disease	CLINICAL PENDING